⚙ FULL DEPLOYMENT SOP (STEP-BY-STEP)

Perfect — this is the **clean, production SOP** you should follow.

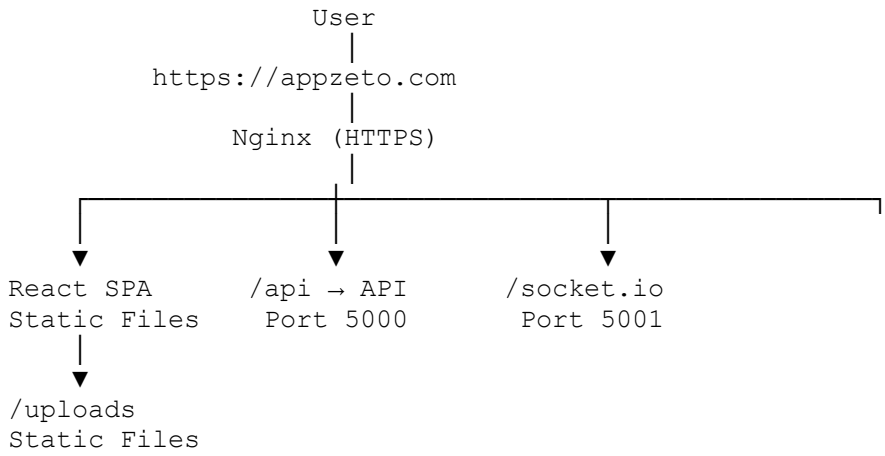
You want:

[Change your domain here only Give and give it to GPT for correct Commands…]

```
Frontend → https://appzeto.com
Backend  → https://appzeto.com/api
```

- Single domain
- Single Nginx config
- Clean + scalable setup

STEP 0: FINAL ARCHITECTURE




```
/api → proxied to backend
```

⚙ STEP 1: PREPARE VPS

```
sudo apt update
sudo apt install nginx git curl -y
```

Set Server Timezone (Recommended)

Set the server timezone to Indian Standard Time (IST):

```
sudo timedatectl set-timezone Asia/Kolkata
```

Verify the timezone:

```
timedatectl | grep "Time zone"
```

Expected output:

```
Time zone: Asia/Kolkata (IST, +0530)
```

Continue with package installation:

```
sudo apt install nginx git curl unzip -y
```

Install Node (if not installed)

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

Install PM2

```
npm install -g pm2
```

⚙ STEP 2: CLONE PROJECT

```
cd ~
git clone <your-repo-url> appzeto
cd appzeto
```

⚙ STEP 3: BACKEND SETUP

```
cd Backend
npm install
```

Create `.env`

```
nano .env
```

Example:

```
# ========================================
# SERVER
# ========================================
PORT=5000
NODE_ENV=production

# ========================================
# DATABASE
# ========================================
MONGODB_URI=

# ========================================
# JWT
# ========================================
JWT_ACCESS_SECRET=ndjdhjhdasdjdhasdjadaskdjasndaskdjadasndaskdjsndaskdjasdkas
nddjkdndkjdnda
JWT_ACCESS_EXPIRES=7d
JWT_REFRESH_SECRET=nsdjkdkddjkjhdhasjdhdassnmadaskdadnkasndasjnkduidsnksdjksd
a
JWT_REFRESH_EXPIRES=30d

# ========================================
# FRONTEND / CORS
# ========================================
FRONTEND_URL=https://appzeto.com

# ========================================
# REDIS & BULLMQ
# ========================================
REDIS_ENABLED=true
REDIS_URL=redis://127.0.0.1:6379
BULLMQ_ENABLED=true

# ========================================
# RATE LIMITING
# ========================================
RATE_LIMIT_ENABLED=true
RATE_LIMIT_WINDOW=15
RATE_LIMIT_MAX=3500
RATE_LIMIT_DEV_MAX=2000
```

```
AUTH_RATE_LIMIT_WINDOW=15
AUTH_RATE_LIMIT_MAX=30

UPLOAD_RATE_LIMIT_WINDOW=15
UPLOAD_RATE_LIMIT_MAX=60

# ========================================
# FILE UPLOADS
# ========================================
UPLOAD_STORAGE_ROOT=/var/www/uploads
UPLOAD_BASE_URL=https://appzeto.com/uploads
UPLOAD_MAX_FILE_SIZE_MB=5

# ========================================
# CLOUDINARY (Optional)
# ========================================
CLOUDINARY_CLOUD_NAME=
CLOUDINARY_API_KEY=
CLOUDINARY_API_SECRET=

# ========================================
# EMAIL
# ========================================
EMAIL_HOST=smtp.gmail.com
EMAIL_PORT=587
EMAIL_USER=
EMAIL_PASS=
EMAIL_FROM="AppZeto <noreply@appzeto.com>"

# ========================================
# GOOGLE MAPS
# ========================================
GOOGLE_MAPS_API_KEY=

# ========================================
# ADMIN
# ========================================
ADMIN_REGISTRATION_CODE=secure-code-to-create-admins
ADMIN_NOTIFICATION_EMAILS=admin1@example.com,admin2@example.com

# ========================================
# RAZORPAY
# ========================================
RAZORPAY_KEY_ID=
RAZORPAY_KEY_SECRET=
RAZORPAY_WEBHOOK_SECRET=

# ========================================
# PASSWORD RESET
# ========================================
PASSWORD_RESET_OTP_EXPIRY_MINUTES=10
PASSWORD_RESET_MAX_ATTEMPTS=5
PASSWORD_RESET_TOKEN_EXPIRY_MINUTES=30

# ========================================
# SOCKET.IO
# ========================================
```

```
SOCKET_CORS_ORIGIN=https://appzeto.com

# ========================================
# FIREBASE
# ========================================
FIREBASE_SERVICE_ACCOUNT_PATH=./config/firebase-service-account.json
FIREBASE_SERVICE_ACCOUNT=

# ========================================
# SMS INDIA HUB
# ========================================
USE_DEFAULT_OTP=false

SMS_INDIA_HUB_USERNAME=APPZETO
SMS_INDIA_HUB_API_KEY=
SMS_INDIA_HUB_SENDER_ID=SMSHUB
SMS_INDIA_HUB_DLT_TEMPLATE_ID=1007801291964877107

# ========================================
# OTP
# ========================================
OTP_EXPIRY_MINUTES=10
OTP_EXPIRY_SECONDS=300
OTP_MAX_ATTEMPTS=5
OTP_RATE_LIMIT=3
OTP_RATE_WINDOW=600
OTP_EXPIRY=5m

# ========================================
# FIREBASE WEB CONFIG
# ========================================
VITE_FIREBASE_API_KEY=
VITE_FIREBASE_AUTH_DOMAIN=
VITE_FIREBASE_PROJECT_ID=
VITE_FIREBASE_STORAGE_BUCKET=
VITE_FIREBASE_MESSAGING_SENDER_ID=
VITE_FIREBASE_APP_ID=
VITE_FIREBASE_MEASUREMENT_ID=
VITE_FIREBASE_DATABASE_URL=
VITE_FIREBASE_VAPID_KEY=
```

Start backend

```
pm2 start server.js --name appzeto-api

pm2 start socket.js --name appzeto-socket

pm2 save
```

⚙ STEP 4: FRONTEND BUILD

```
cd ../Frontend
npm install
```

Set production env

```
nano .env
VITE_API_BASE_URL=/api
```

Build

```
npm run build
```

Move build to Nginx folder

```
sudo mkdir -p /var/www/appzeto
sudo rm -rf /var/www/appzeto/*
sudo cp -r dist/* /var/www/appzeto/
```

⚙ STEP 5: NGINX CONFIG (MOST IMPORTANT)

Create config

```
sudo nano /etc/nginx/sites-available/appzeto.conf
```

□ FULL CONFIG (FINAL)

```
server {
    server_name appzeto.com www.appzeto.com;

    root /var/www/appzeto;
    index index.html;

    # Maximum upload size
    client_max_body_size 6M;

    # Keep client connections alive
    keepalive_timeout 65;
```

```
# Enable Gzip Compression
gzip on;
gzip_comp_level 5;
gzip_min_length 1024;
gzip_proxied any;
gzip_types
    text/plain
    text/css
    application/json
    application/javascript
    text/xml
    application/xml
    application/xml+rss
    image/svg+xml;


# ============================
# Frontend (React)
# ============================
location / {
    try_files $uri $uri/ /index.html;
}

# ============================
# Static Uploads
# ============================
location /uploads/ {
    alias /var/www/uploads/;
    expires 30d;
    add_header Cache-Control "public, max-age=2592000, immutable";
    add_header Cross-Origin-Resource-Policy "cross-origin" always;
    access_log off;
    try_files $uri =404;
}

# ============================
# Health Check
# ============================
location /health {
    proxy_pass http://127.0.0.1:5000;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}

# ============================
# Readiness Check
# ============================
location /ready {
    proxy_pass http://127.0.0.1:5000;
    proxy_http_version 1.1;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
```

```
}

# ==========================
# Backend API
# ==========================
location /api/ {
    limit_req zone=api_limit burst=60 nodelay;
    limit_conn perip 30;

    proxy_pass http://127.0.0.1:5000;
    proxy_http_version 1.1;

    proxy_set_header Connection "";
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;

    proxy_connect_timeout 10s;
    proxy_send_timeout 60s;
    proxy_read_timeout 60s;

    proxy_buffering on;
    proxy_buffers 16 16k;
    proxy_busy_buffers_size 64k;
}

# ==========================
# Socket.IO Server
# ==========================
location /socket.io/ {
    proxy_pass http://127.0.0.1:5001;
    proxy_http_version 1.1;

    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection "upgrade";
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;

    proxy_read_timeout 75s;
    proxy_send_timeout 75s;
    proxy_cache_bypass $http_upgrade;
}

# ==========================
# SSL Configuration
# ==========================
listen [::]:443 ssl ipv6only=on;
listen 443 ssl;

ssl_certificate /etc/letsencrypt/live/appzeto.com/fullchain.pem;
ssl_certificate_key /etc/letsencrypt/live/appzeto.com/privkey.pem;

include /etc/letsencrypt/options-ssl-nginx.conf;
ssl_dhparam /etc/letsencrypt/ssl-dhparams.pem;
```

```
}
```

Enable config

```
sudo ln -s /etc/nginx/sites-available/appzeto.conf /etc/nginx/sites-enabled/
```

Remove default (optional)

```
sudo rm /etc/nginx/sites-enabled/default
```

Restart nginx

```
sudo nginx -t
sudo systemctl restart nginx
```

✅ Upload files to server

Upload Directory

Add

```
sudo mkdir -p /var/www/uploads

sudo chown -R www-data:www-data /var/www/uploads

sudo chmod -R 755 /var/www/uploads
```

Explain

- Backend stores uploads here.
- Nginx serves uploads directly.
- Uploads are available at:

```
https://appzeto.com/uploads/file.jpg
```

⚙ STEP 6: DOMAIN SETUP

In DNS:

```
Type: A
Name: @
Value: YOUR_VPS_IP

Type: A
Name: www
Value: YOUR_VPS_IP
```

⚙ STEP 7: SSL (HTTPS)

```
sudo apt install certbot python3-certbot-nginx -y
sudo certbot --nginx -d appzeto.com -d www.appzeto.com
```

 Choose:

```
2 → Redirect HTTP to HTTPS
```

⚙ STEP 8: AUTO DEPLOY SCRIPT (IMPORTANT)

Create script

```
nano ~/deploy.sh
```

Paste:

```
#!/bin/bash

echo " Deploying..."

cd ~/appzeto

git pull origin main

# FRONTEND
cd Frontend
npm install
npm run build
sudo rm -rf /var/www/appzeto/*
sudo cp -r dist/* /var/www/appzeto/
```

```
# BACKEND
cd ../Backend
npm install

pm2 restart appzeto-api

pm2 restart appzeto-socket

echo "✅ Done"
```

Make executable

```
chmod +x ~/deploy.sh
```

Deploy anytime

```
./deploy.sh
```

□ TEST

Frontend

https://appzeto.com

API

https://appzeto.com/api

Health

https://appzeto.com/health

Ready

https://appzeto.com/ready

Uploads

https://appzeto.com/uploads/test.jpg

Socket.IO

wss://appzeto.com/socket.io/

⚙ IMPORTANT RULES

✅Frontend must use:

```
VITE_API_BASE_URL=/api
```

❌NEVER use:

```
http://localhost:5000 ❌
```

✅Backend runs on:

```
http://127.0.0.1:5000
```

Backend API

127.0.0.1:5000

Socket.IO

127.0.0.1:5001

□ FINAL RESULT

✔Single domain

✔Clean API routing

✔No CORS issues

✔Easy deploy

✔Production-ready

□ PART 2: DOMAIN POINTING (HOSTINGER)

□ Go to **Hostinger DNS panel**

✓Add A Records

Type	Name	Value
A	@	YOUR_VPS_IP
A	www	YOUR_VPS_IP

⚙ PART 3: DAILY WORKFLOW (VERY IMPORTANT)

WHEN YOU PUSH CODE

 You DON’T need to touch domain or nginx again

✅Just run:

```
cd ~
./deploy.sh
```

---

deploy.sh (FINAL VERSION)

```
#!/bin/bash

echo " Deploy started..."

cd ~/appzeto

git pull origin main

# FRONTEND
cd Frontend
npm install
npm run build
sudo rm -rf /var/www/appzeto/*
sudo cp -r dist/* /var/www/appzeto/
```

```
# BACKEND
cd ../Backend
npm install
pm2 restart appzeto-api

pm2 restart appzeto-socket

echo "✅ Deploy finished"
```

```
Push → GitHub → VPS → deploy.sh auto runs
```

□ PART 6: MAINTENANCE TASKS

✅1. Check logs

```
pm2 logs appzeto-backend
```

✅2. Restart backend

```
pm2 restart appzeto-api
```

✅3. Restart nginx

```
sudo systemctl restart nginx
```

✅4. Check nginx errors

```
sudo tail -f /var/log/nginx/error.log
```

✅5. Check running ports

```
443 → HTTPS

80 → HTTP Redirect

5000 → API
```

```
5001 → Socket.IO
```

✅6. Backup (important)

```
cp -r ~/appzeto ~/backup/appzeto-$(date +%F)
```

---

2. Restart socket backedn server

```
pm2 restart appzeto-socket
```

⚙ PART 7: COMMON MISTAKES

❌API not working

 Fix frontend env:

```
VITE_API_BASE_URL=/api
```

❌Nginx error

```
sudo nginx -t
```

❌Backend not running

```
pm2 list
```

❌Blank page

 Build missing:

```
npm run build
```

Socket not connecting

```
pm2 logs appzeto-socket
```

Check

```
Port 5001
```

Uploads returning 404

Check

```
/var/www/uploads
```

Permissions

```
www-data:www-data
```

Health endpoint failing

```
curl https://appzeto.com/health
```

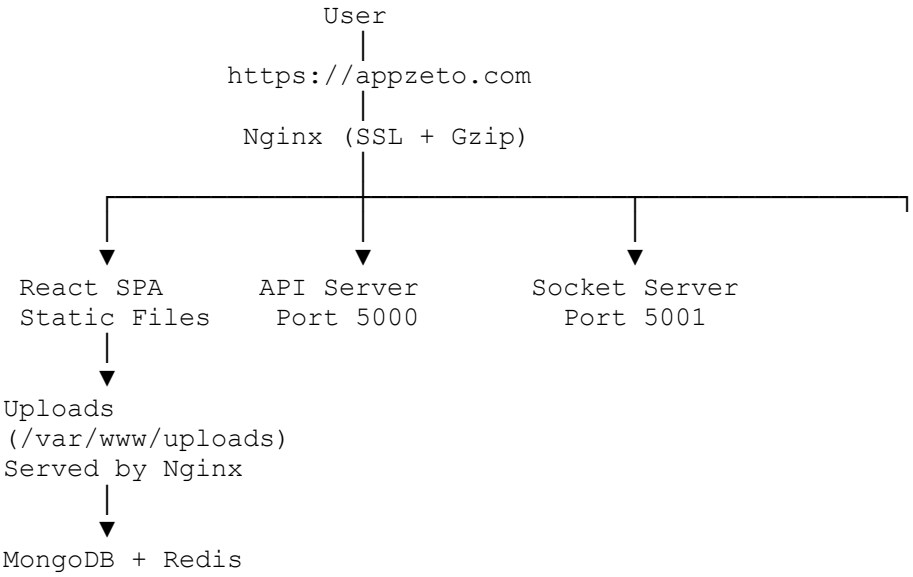
□ PART 8: WHEN TO TOUCH WHAT

Action	Required
Push new code	run deploy.sh
Change domain	update DNS
Change API route	update nginx
Backend crash	pm2 restart
SSL expire	certbot renew

Upload issue Check `/var/www/uploads`

Action	Required
Socket issue	Restart Socket.IO PM2 process
Health failing	Check API server
Ready failing	Check backend readiness

□ FINAL FLOW (VERY IMPORTANT)



□ FINAL RESULT

✔One Domain

✔HTTPS Enabled

✔React SPA

✔Backend API (Port 5000)

✔Dedicated Socket.IO Server (Port 5001)

✔Static Uploads Served by Nginx

✔Health & Ready Endpoints

✔Gzip Compression

✔Browser Caching

✔API Rate Limiting

✔Connection Limiting

✔SSL Auto Renewal

✔PM2 Process Management

✔Production Ready

✔Scalable Architecture